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Last login: Sun Jun 28 06:26:52 on ttys000
stephenrajva@Stephens-MacBook-Air-4 ~ % cd "/Users/stephenrajva/Downloads/HPC-Up
load 2/SST_Stimulator/MiniFE/Mesh/PPN_2"
python3 analyze_universal.py
!! missing: ./results/results/mesh_2x2_ppn2_nodes8_stats.csv (skipping)
!! missing: ./results/results/mesh_4x4_ppn2_nodes32_stats.csv (skipping)
!! missing: ./results/results/mesh_8x8_ppn2_nodes128_stats.csv (skipping)
!! missing: ./results/results/mesh_16x16_ppn2_nodes512_stats.csv (skipping)
!! missing: ./results/results/mesh_32x32_ppn2_nodes2048_stats.csv (skipping)

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MiniFE_Mesh_PPN2

Size	Latency(ns)	Hops	Thru(GB/s)	Packets	Bits	Idle%	Stalls
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Saved: MiniFE_Mesh_PPN2_latency_ns.png

Saved: MiniFE_Mesh_PPN2_hops.png

Saved: MiniFE_Mesh_PPN2_throughput.png

Saved: MiniFE_Mesh_PPN2_packets.png

Saved: MiniFE_Mesh_PPN2_bits.png

Saved: MiniFE_Mesh_PPN2_idle_pct.png

Saved: MiniFE_Mesh_PPN2_stalls.png

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stephenrajva@Stephens-MacBook-Air-4 PPN_2 % cd "/Users/stephenrajva/Downloads/HP
C-Uplod 2/SST_Stimulator/MiniFE/Mesh/PPN_2"

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grep -n "results\|os.path.join\|RESULTS\|csv" analyze_universal.py | head -30

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14:import os, csv

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20:DATA_DIR = os.environ.get("DATA_DIR", "./results")

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26:# (x-axis label, dims tuple, csv filename)

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28:     ("2x2", (2, 2), "results/mesh_2x2_ppn2_nodes8_stats.csv"),

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29:     ("4x4", (4, 4), "results/mesh_4x4_ppn2_nodes32_stats.csv"),

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30:     ("8x8", (8, 8), "results/mesh_8x8_ppn2_nodes128_stats.csv"),

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31:     ("16x16", (16, 16), "results/mesh_16x16_ppn2_nodes512_stats.csv"),

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32:     ("32x32", (32, 32), "results/mesh_32x32_ppn2_nodes2048_stats.csv"),

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37:def read_csv_rows(path):

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39:     reader = csv.reader(f)

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50:def metrics_from_csv(path):

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52:     idx, rows = read_csv_rows(path)

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125:     path = os.path.join(DATA_DIR, fname)

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128:     m = metrics_from_csv(path)

```

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stephenrajva@Stephens-MacBook-Air-4 PPN_2 % cd "/Users/stephenrajva/Downloads/HP
C-Uplod 2/SST_Stimulator/MiniFE/Mesh/PPN_2"

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python3 - << 'EOF'

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import re

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f = "analyze_universal.py"

```

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src = open(f).read()

```

```

runs = '''RUNS = [

```

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    ("2x2", (2, 2), "mesh_2x2_ppn2_nodes8_stats.csv"),

```

```

    ("4x4", (4, 4), "mesh_4x4_ppn2_nodes32_stats.csv"),

```

```

    ("8x8", (8, 8), "mesh_8x8_ppn2_nodes128_stats.csv"),

```

```

    ("16x16", (16, 16), "mesh_16x16_ppn2_nodes512_stats.csv"),

```

```

    ("32x32", (32, 32), "mesh_32x32_ppn2_nodes2048_stats.csv"),

```

```

]'''

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src = re.sub(r'^RUNS\s*=\s*\[.*?\]', runs, src, count=1, flags=re.M | re.S)

```

```

open(f, "w").write(src)

```

```

print("=== mesh RUNS now (bare filenames) ===")

```

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for line in src.splitlines():

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    if 'ppn2_nodes' in line:

```

```

        print(line)

```

```

EOF

```

```

python3 analyze_universal.py

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=== mesh RUNS now (bare filenames) ===

```
("2x2", (2, 2), "mesh_2x2_ppn2_nodes8_stats.csv"),
("4x4", (4, 4), "mesh_4x4_ppn2_nodes32_stats.csv"),
("8x8", (8, 8), "mesh_8x8_ppn2_nodes128_stats.csv"),
("16x16", (16, 16), "mesh_16x16_ppn2_nodes512_stats.csv"),
("32x32", (32, 32), "mesh_32x32_ppn2_nodes2048_stats.csv"),
```

MiniFE_Mesh_PPN2

Size	Latency(ns)	Hops	Thru(GB/s)	Packets	Bits	Idle%	Stalls
2x2	198.00	1.33	0.3247	6184	3588096	90.5	0
4x4	294.83	2.67	1.0479	40152	18551552	95.7	0
8x8	467.01	5.33	4.8257	268648	129776256	97.5	0
16x16	810.37	10.67	32.0120	1925832	1320512384	98.3	0
32x32	1952.97	21.33	323.5003	15521416	22042224000	98.5	0

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Saved: MiniFE_Mesh_PPN2_bits.png

Saved: MiniFE_Mesh_PPN2_idle_pct.png

Saved: MiniFE_Mesh_PPN2_stalls.png

stephenrajva@Stephens-MacBook-Air-4 PPN_2 % cd "/Users/stephenrajva/Downloads/HP C-Upload 2/SST_Stimulator/MiniFE/Torus/PPN_2"

python3 analyze_universal.py

MiniFE_Torus_PPN2

Size	Latency(ns)	Hops	Thru(GB/s)	Packets	Bits	Idle%	Stalls
2x2	198.00	1.33	0.3247	6184	3588096	92.9	0
4x4	286.44	2.13	0.9961	38904	17442496	96.2	0
8x8	451.13	4.06	4.7122	258904	122329920	97.6	0
16x16	798.32	8.03	31.5980	1869720	1234056256	98.3	0
32x32	2064.99	16.02	322.8922	15119896	20018771520	98.5	0

Saved: MiniFE_Torus_PPN2_latency_ns.png

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Saved: MiniFE_Torus_PPN2_bits.png

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Saved: MiniFE_Torus_PPN2_stalls.png

stephenrajva@Stephens-MacBook-Air-4 PPN_2 % cd "/Users/stephenrajva/Downloads/HP C-Upload 2/SST_Stimulator/MiniFE/Mesh/PPN_2"

grep -n "PPN = 1\|PPN=1\|PPN = {\|set_title\|suptitle\|title(" analyze_universal.py

3:# analyze_universal.py - PPN=1 scaling graphs for ANY topology + app

168: ax.set_title(f"{STUDY}: {short} vs size (PPN = 1)")

stephenrajva@Stephens-MacBook-Air-4 PPN_2 % cd "/Users/stephenrajva/Downloads/HP C-Upload 2/SST_Stimulator/MiniFE/Mesh/PPN_2"

sed -i 's/PPN = 1/PPN = 2/g; s/PPN=1/PPN=2/g' analyze_universal.py

python3 analyze_universal.py

MiniFE_Mesh_PPN2

Size	Latency(ns)	Hops	Thru(GB/s)	Packets	Bits	Idle%	Stalls
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16x16    810.37      10.67    32.0120    1925832    1320512384  98.3    0
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stephenrajva@Stephens-MacBook-Air-4 PPN_2 % cd "/Users/stephenrajva/Downloads/HP
C-Upload 2/SST_Stimulator/MiniFE/Mesh/PPN_2"
sed -i '' 's/PPN = 1/PPN = 2/g; s/PPN=1/PPN=2/g' analyze_universal.py
python3 analyze_universal.py

```

MiniFE_Mesh_PPN2

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Saved: MiniFE_Mesh_PPN2_stalls.png
stephenrajva@Stephens-MacBook-Air-4 PPN_2 % cd "/Users/stephenrajva/Downloads/HP
C-Upload 2/SST_Stimulator/MiniFE/Mesh/PPN_2"
ls -lt *.png
pwd
-rw-r--r--@ 1 stephenrajva  staff   33196  28 Jun 06:40 MiniFE_Mesh_PPN2_stalls.png
-rw-r--r--@ 1 stephenrajva  staff   53694  28 Jun 06:40 MiniFE_Mesh_PPN2_idle_pct.png
-rw-r--r--  1 stephenrajva  staff   52892  28 Jun 06:40 MiniFE_Mesh_PPN2_bits.png
-rw-r--r--@ 1 stephenrajva  staff   53567  28 Jun 06:40 MiniFE_Mesh_PPN2_packets.png
-rw-r--r--@ 1 stephenrajva  staff   53614  28 Jun 06:40 MiniFE_Mesh_PPN2_throughput.png
-rw-r--r--@ 1 stephenrajva  staff   55269  28 Jun 06:40 MiniFE_Mesh_PPN2_hops.png
-rw-r--r--@ 1 stephenrajva  staff   59424  28 Jun 06:40 MiniFE_Mesh_PPN2_latency_ns.png
/Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniFE/Mesh/PPN_2
stephenrajva@Stephens-MacBook-Air-4 PPN_2 %

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